

SEMINAR 4 (Pilot). LIMITS OF SEQUENCES

1. Compute

(a) $\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots + \frac{1}{n(n+1)(n+2)} \right);$ (b) (HW) $\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3n-2)(3n+1)} \right).$

2. Find the following limits:

(a) $\lim_{n \rightarrow \infty} \frac{n}{n^3 + 1} \cdot \cos(n+7);$ (b) (HW) $\lim_{n \rightarrow \infty} \left(\sqrt{n^2 + 1} - \sqrt{n^2 - 1} \right) \cdot \sin(n^2 + 3);$
(c) $\lim_{n \rightarrow \infty} \frac{(3n-7)^4(2n + \sin n - 1)^2}{(5n^2 + n - 2)^3};$ (d) (HW) $\lim_{n \rightarrow \infty} \frac{(4 \cos n - 3n)^2(2n^5 - n^3 + 1)}{(6n^3 + 5n \sin n)(n+2)^4}.$

3. Find the following limits:

(a) $\lim_{n \rightarrow \infty} \left(\frac{n^2 + 2022}{n^2 + 2024} \right)^{n^2 + 2026};$ (b) $\lim_{n \rightarrow +\infty} (8n+5) \left(\ln(4n-2) - \ln(4n+1) \right).$

4. Find the following limits:

(a) $\lim_{n \rightarrow \infty} \left(\frac{n+4}{n+1} \right)^{n+2};$ (b) $\lim_{n \rightarrow +\infty} (3n+1) \left(\ln(2n+7) - \ln(2n+5) \right).$

5. Prove that the sequence $\{x_n\}$, given by $x_{n+1} = \sqrt{2x_n}$, $x_1 = \sqrt{2}$, is convergent and find the limit of this sequence.

6. (HW) Prove that the sequence $\{x_n\}$, given by $x_{n+1} = \sqrt{2+x_n}$, $x_1 = \sqrt{2}$, is convergent and find the limit of this sequence.

7. Prove that the sequence $\{x_n\}$, given by $x_{n+1} = \sqrt{15+2x_n}$, is convergent and find the limit of this sequence if

(a) $x_1 = 1;$ (b) (HW) $x_1 = 7.$

8. Prove that the sequence $\{x_n\}$, given by $x_{n+1} = \frac{4}{3}x_n - x_n^2$, is convergent and find the limit of this sequence if $x_1 = \frac{1}{2}.$

9. (HW) Prove that the sequence $\{x_n\}$, given by $x_{n+1} = \frac{1}{2} \left(x_n + \frac{1}{x_n} \right)$, is convergent and find the limit of this sequence if $x_0 = 2.$ (*Hint.* Use inequality $\frac{x+y}{2} \geq \sqrt{xy}$ for $x \geq 0, y \geq 0$)

10*. (HW) Compute the limit

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \dots + \frac{2n-1}{2^n} \right).$$